CelebA - eleven-method native validation comparison

Main table | 10 shared seeds: 91001-91010 | mean +/- sample SD (ddof=1)

Best: bold. Second-best: underlined. Fairness ranking requires mean ACC >80%; dagger marks ineligible values. Displayed ties share rank.

IID (alpha=5000)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	84.72 ± 11.74	0.0384 ± 0.0176	0.0937 ± 0.0366	89.57 ± 0.78	0.0519 ± 0.0096	0.1212 ± 0.0064	68.19 ± 5.77	0.1004 ± 0.0410†	0.0993 ± 0.0380†	66.01 ± 3.57	0.0950 ± 0.0477†	0.0807 ± 0.0413†	84.37 ± 2.38	0.0280 ± 0.0076	0.0693 ± 0.0161
FairFed	89.47 ± 0.85	0.0414 ± 0.0076	0.1102 ± 0.0077	89.79 ± 0.76	0.0489 ± 0.0042	0.1188 ± 0.0043	69.47 ± 4.65	0.1158 ± 0.0635†	0.1120 ± 0.0554†	62.59 ± 5.11	0.1594 ± 0.0822†	0.1461 ± 0.0668†	85.55 ± 2.31	0.0288 ± 0.0081	0.0781 ± 0.0116
Median	86.28 ± 1.31	0.0510 ± 0.0081	0.0980 ± 0.0117	86.52 ± 1.35	0.0610 ± 0.0087	0.1062 ± 0.0111	84.66 ± 1.47	0.0438 ± 0.0135	0.0825 ± 0.0142	85.00 ± 1.38	0.0484 ± 0.0090	0.0852 ± 0.0127	85.54 ± 1.50	0.0534 ± 0.0096	0.0965 ± 0.0124
FLTrust	89.43 ± 0.75	0.0432 ± 0.0075	0.1111 ± 0.0062	89.67 ± 0.82	0.0518 ± 0.0072	0.1207 ± 0.0064	**89.68 ± 0.74**	0.0422 ± 0.0069	0.1119 ± 0.0071	**89.68 ± 0.74**	0.0422 ± 0.0069	0.1119 ± 0.0071	**89.72 ± 0.64**	0.0496 ± 0.0061	0.1177 ± 0.0057
FedAA-DDPG	88.35 ± 1.33	0.0455 ± 0.0116	0.1083 ± 0.0100	89.06 ± 0.92	0.0442 ± 0.0137	0.1097 ± 0.0143	88.80 ± 1.06	0.0423 ± 0.0116	0.1071 ± 0.0115	88.82 ± 1.21	0.0385 ± 0.0131	0.1042 ± 0.0152	88.60 ± 1.12	0.0493 ± 0.0089	0.1136 ± 0.0095
LASA	89.39 ± 1.42	0.0377 ± 0.0049	0.1074 ± 0.0088	89.46 ± 0.79	0.0493 ± 0.0066	0.1176 ± 0.0059	85.75 ± 1.71	0.0330 ± 0.0109	0.0811 ± 0.0138	85.96 ± 1.62	0.0366 ± 0.0105	0.0839 ± 0.0152	88.20 ± 1.40	0.0450 ± 0.0073	0.1050 ± 0.0113
FairGuard	89.18 ± 0.83	0.0317 ± 0.0026	0.0998 ± 0.0062	89.57 ± 1.24	0.0369 ± 0.0071	0.1078 ± 0.0092	49.32 ± 1.59	0.0042 ± 0.0132†	0.0049 ± 0.0156†	50.33 ± 1.72	0.0000 ± 0.0000†	0.0000 ± 0.0000†	83.20 ± 1.76	0.0205 ± 0.0098	0.0491 ± 0.0199
FLTrust+FairGuard	**89.93 ± 1.10**	0.0365 ± 0.0075	0.1074 ± 0.0113	**90.37 ± 0.60**	0.0363 ± 0.0086	0.1118 ± 0.0100	77.83 ± 12.15	0.0298 ± 0.0299†	0.0607 ± 0.0354†	75.99 ± 16.52	0.0293 ± 0.0146†	0.0652 ± 0.0419†	86.44 ± 5.76	0.0368 ± 0.0144	0.0903 ± 0.0325
GuardFed-AD2+	88.26 ± 1.04	**0.0097 ± 0.0075**	0.0625 ± 0.0138	88.39 ± 0.63	**0.0107 ± 0.0096**	0.0607 ± 0.0103	88.47 ± 0.91	**0.0061 ± 0.0042**	**0.0640 ± 0.0142**	88.69 ± 0.81	**0.0077 ± 0.0046**	**0.0638 ± 0.0070**	88.38 ± 0.89	**0.0165 ± 0.0072**	0.0559 ± 0.0156
LoGoFair-DP	83.64 ± 11.34	0.0453 ± 0.0189	**0.0071 ± 0.0059**	88.03 ± 0.79	0.0564 ± 0.0077	**0.0079 ± 0.0040**	69.18 ± 5.69	0.0161 ± 0.0115†	0.0103 ± 0.0071†	66.76 ± 3.39	0.0129 ± 0.0106†	0.0100 ± 0.0055†	83.61 ± 2.26	0.0284 ± 0.0115	**0.0077 ± 0.0061**
FLGMM	89.33 ± 0.99	0.0404 ± 0.0065	0.1074 ± 0.0073	89.72 ± 0.83	0.0458 ± 0.0078	0.1171 ± 0.0049	89.34 ± 1.01	0.0424 ± 0.0070	0.1113 ± 0.0081	89.34 ± 1.01	0.0423 ± 0.0075	0.1113 ± 0.0080	89.49 ± 0.93	0.0460 ± 0.0061	0.1143 ± 0.0063

non-IID (alpha=5)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	86.85 ± 5.45	0.0428 ± 0.0137	0.0958 ± 0.0353	89.51 ± 1.82	0.0492 ± 0.0045	0.1161 ± 0.0143	74.76 ± 13.38	0.0644 ± 0.1097†	0.0614 ± 0.0893†	74.51 ± 13.37	0.0522 ± 0.0913†	0.0591 ± 0.0756†	85.48 ± 2.58	0.0267 ± 0.0157	0.0741 ± 0.0247
FairFed	89.59 ± 0.97	0.0388 ± 0.0057	0.1092 ± 0.0071	89.83 ± 0.92	0.0462 ± 0.0081	0.1163 ± 0.0071	71.45 ± 11.85	0.0759 ± 0.0860†	0.0730 ± 0.0885†	67.06 ± 11.91	0.1307 ± 0.1184†	0.1171 ± 0.1067†	86.18 ± 2.41	0.0328 ± 0.0133	0.0829 ± 0.0204
Median	86.22 ± 1.31	0.0518 ± 0.0068	0.0963 ± 0.0096	86.51 ± 1.24	0.0594 ± 0.0084	0.1067 ± 0.0097	84.47 ± 1.30	0.0425 ± 0.0118	0.0799 ± 0.0113	84.48 ± 1.35	0.0461 ± 0.0142	0.0839 ± 0.0124	85.47 ± 1.26	0.0502 ± 0.0111	0.0929 ± 0.0086
FLTrust	89.68 ± 0.60	0.0445 ± 0.0049	0.1132 ± 0.0060	89.83 ± 0.69	0.0493 ± 0.0054	0.1197 ± 0.0044	**89.48 ± 0.70**	0.0441 ± 0.0063	0.1120 ± 0.0041	**89.48 ± 0.70**	0.0441 ± 0.0063	0.1120 ± 0.0041	**89.64 ± 0.71**	0.0475 ± 0.0052	0.1153 ± 0.0053
FedAA-DDPG	88.60 ± 1.14	0.0456 ± 0.0125	0.1069 ± 0.0151	88.56 ± 1.19	0.0491 ± 0.0132	0.1110 ± 0.0103	88.53 ± 1.31	0.0421 ± 0.0231	0.1031 ± 0.0212	88.52 ± 1.60	0.0409 ± 0.0122	0.1020 ± 0.0149	88.39 ± 1.14	0.0597 ± 0.0189	0.1188 ± 0.0190
LASA	89.27 ± 0.85	0.0386 ± 0.0057	0.1070 ± 0.0084	89.51 ± 0.88	0.0500 ± 0.0096	0.1181 ± 0.0059	85.91 ± 1.69	0.0301 ± 0.0069	0.0767 ± 0.0129	85.93 ± 1.64	0.0345 ± 0.0113	0.0791 ± 0.0128	88.15 ± 1.22	0.0417 ± 0.0037	0.1026 ± 0.0074
FairGuard	89.39 ± 1.11	0.0314 ± 0.0054	0.1026 ± 0.0086	89.84 ± 0.76	0.0394 ± 0.0103	0.1109 ± 0.0090	59.95 ± 16.51	0.0066 ± 0.0094†	0.0169 ± 0.0251†	61.29 ± 15.59	0.0108 ± 0.0161†	0.0206 ± 0.0317†	78.37 ± 14.94	0.0209 ± 0.0104†	0.0552 ± 0.0238†
FLTrust+FairGuard	**89.69 ± 1.68**	0.0357 ± 0.0055	0.1085 ± 0.0102	**90.07 ± 1.34**	0.0426 ± 0.0062	0.1144 ± 0.0088	81.90 ± 11.66	0.0400 ± 0.0214	0.0851 ± 0.0350	80.44 ± 12.06	0.0359 ± 0.0262	0.0763 ± 0.0174	79.69 ± 13.94	0.0408 ± 0.0124†	0.0892 ± 0.0276†
GuardFed-AD2+	88.59 ± 1.17	**0.0070 ± 0.0042**	0.0651 ± 0.0090	88.44 ± 0.74	**0.0100 ± 0.0070**	0.0589 ± 0.0092	88.41 ± 0.92	**0.0078 ± 0.0056**	**0.0616 ± 0.0141**	88.22 ± 0.78	**0.0126 ± 0.0093**	**0.0584 ± 0.0189**	88.34 ± 0.96	**0.0087 ± 0.0066**	0.0596 ± 0.0146
LoGoFair-DP	85.64 ± 4.92	0.0445 ± 0.0232	**0.0055 ± 0.0051**	87.96 ± 1.66	0.0535 ± 0.0147	**0.0060 ± 0.0036**	74.94 ± 12.90	0.0192 ± 0.0147†	0.0084 ± 0.0054†	74.53 ± 12.82	0.0196 ± 0.0178†	0.0090 ± 0.0061†	84.53 ± 2.27	0.0358 ± 0.0122	**0.0051 ± 0.0045**
FLGMM	89.02 ± 0.91	0.0396 ± 0.0066	0.1071 ± 0.0078	89.28 ± 0.98	0.0525 ± 0.0077	0.1185 ± 0.0068	89.15 ± 0.88	0.0428 ± 0.0049	0.1088 ± 0.0055	89.17 ± 1.15	0.0423 ± 0.0048	0.1083 ± 0.0105	**89.64 ± 0.72**	0.0446 ± 0.0084	0.1147 ± 0.0053

Reading: Higher ACC and lower AEOD/ASPD are better. Best/second-best eligible displayed means are **bold**/underlined; displayed ties share rank. † retains a disparity whose corresponding mean ACC is <=80%, excluding it only from fairness ranking. ACC ranking and all seeds/means/sample SDs are unchanged. Eligibility uses unrounded mean ACC; SD does not rank. This post-hoc display rule was author-approved on 2026-10-12, not preregistered. Marks do not establish significance or undefined metrics. ACC is percent; gaps are [0,1]; AEOD is the absolute TPR gap, not full equalized odds. Shading identifies GuardFed.

Method scope: Native predictions preserve each method's own postprocessing; this does not isolate aggregation effects. FairFed/FairGuard/composition, FedAA and LASA are documented adaptations. LoGoFair-DP uses fitted demographic-parity postprocessing, 20 image-ID virtual cohorts (not training clients), fit seed 1719 and 30 postprocessing rounds. FLGMM uses the author-code full-model/largest-cluster adaptation and native uncalibrated outputs.

Coverage and provenance: Eleven methods, two distributions, five conditions and 10 seeds give 1100 evaluation records, not 1100 independent training models: LoGoFair reuses FedAvg checkpoints. Backbone checkpoints are terminal round 70. FLGMM contributes 96 new and 4 explicitly reused screen records; its selected recipe is warmup 20/control width 2.0/local learning rate 0.001. Original mixed training/replay environments and separate CPU saved-state verification remain disclosed; no cross-runtime equivalence is claimed.

Selection boundary: Seed 91001 participated in validation selection; other validation seeds and historical test metadata were exposed. The 9/6-seed panels are sensitivity views, not untouched test cohorts. No score ranking, significance, final-test or complete 17-method claim is made. This display does not select the manuscript's final primary endpoint.

Source: root-adopted eleven-method native1100 display table; no new statistics, fits, inference or training.

CelebA - eleven-method native validation comparison

Appendix sensitivity analysis | 9 shared seeds: 91002-91010 | mean +/- sample SD (ddof=1)

Best: bold. Second-best: underlined. Fairness ranking requires mean ACC >80%; dagger marks ineligible values. Displayed ties share rank.

IID (alpha=5000)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	88.39 ± 1.84	0.0426 ± 0.0119	0.1041 ± 0.0171	89.72 ± 0.65	0.0522 ± 0.0101	0.1225 ± 0.0053	69.08 ± 5.35	0.0939 ± 0.0376†	0.0962 ± 0.0389†	66.61 ± 3.21	0.0904 ± 0.0481†	0.0781 ± 0.0429†	84.91 ± 1.77	0.0289 ± 0.0075	0.0702 ± 0.0168
FairFed	89.61 ± 0.78	0.0411 ± 0.0080	0.1108 ± 0.0078	89.91 ± 0.71	0.0483 ± 0.0040	0.1189 ± 0.0046	70.47 ± 3.62	0.1030 ± 0.0521†	0.1009 ± 0.0454†	63.49 ± 4.50	0.1435 ± 0.0691†	0.1346 ± 0.0595†	86.17 ± 1.29	0.0301 ± 0.0076	0.0797 ± 0.0111
Median	86.51 ± 1.16	0.0509 ± 0.0086	0.0988 ± 0.0121	86.71 ± 1.30	0.0610 ± 0.0093	0.1070 ± 0.0114	84.87 ± 1.39	0.0433 ± 0.0142	0.0830 ± 0.0150	85.25 ± 1.20	0.0483 ± 0.0096	0.0860 ± 0.0132	85.73 ± 1.47	0.0532 ± 0.0101	0.0974 ± 0.0128
FLTrust	89.47 ± 0.78	0.0420 ± 0.0069	0.1108 ± 0.0065	89.73 ± 0.85	0.0521 ± 0.0075	0.1214 ± 0.0064	**89.73 ± 0.76**	0.0416 ± 0.0071	0.1124 ± 0.0074	**89.73 ± 0.76**	0.0416 ± 0.0071	0.1124 ± 0.0074	**89.79 ± 0.64**	0.0494 ± 0.0064	0.1181 ± 0.0059
FedAA-DDPG	88.58 ± 1.18	0.0446 ± 0.0120	0.1093 ± 0.0101	89.24 ± 0.77	0.0455 ± 0.0139	0.1124 ± 0.0121	88.85 ± 1.11	0.0428 ± 0.0122	0.1076 ± 0.0121	88.90 ± 1.26	0.0363 ± 0.0116	0.1032 ± 0.0158	88.86 ± 0.81	0.0478 ± 0.0078	0.1136 ± 0.0101
LASA	89.77 ± 0.81	0.0373 ± 0.0050	0.1099 ± 0.0042	89.59 ± 0.72	0.0489 ± 0.0069	0.1184 ± 0.0057	86.15 ± 1.21	0.0315 ± 0.0103	0.0810 ± 0.0146	86.15 ± 1.61	0.0379 ± 0.0103	0.0861 ± 0.0143	88.38 ± 1.35	0.0446 ± 0.0077	0.1059 ± 0.0116
FairGuard	89.27 ± 0.84	0.0316 ± 0.0028	0.1002 ± 0.0064	89.65 ± 1.29	0.0369 ± 0.0075	0.1085 ± 0.0094	49.42 ± 1.64	0.0046 ± 0.0139†	0.0055 ± 0.0164†	50.56 ± 1.67	0.0000 ± 0.0000†	0.0000 ± 0.0000†	83.20 ± 1.87	0.0185 ± 0.0080	0.0471 ± 0.0199
FLTrust+FairGuard	**90.22 ± 0.65**	0.0371 ± 0.0076	0.1101 ± 0.0078	**90.44 ± 0.59**	0.0362 ± 0.0091	0.1131 ± 0.0098	76.75 ± 12.37	0.0308 ± 0.0315†	0.0588 ± 0.0369†	74.59 ± 16.88	0.0298 ± 0.0154†	0.0627 ± 0.0437†	86.38 ± 6.11	0.0367 ± 0.0153	0.0903 ± 0.0345
GuardFed-AD2+	88.35 ± 1.06	**0.0097 ± 0.0080**	0.0623 ± 0.0146	88.41 ± 0.66	**0.0115 ± 0.0098**	0.0604 ± 0.0108	88.62 ± 0.82	**0.0054 ± 0.0039**	**0.0667 ± 0.0120**	88.84 ± 0.68	**0.0068 ± 0.0038**	**0.0633 ± 0.0072**	88.55 ± 0.77	**0.0171 ± 0.0074**	0.0555 ± 0.0165
LoGoFair-DP	87.19 ± 1.67	0.0503 ± 0.0107	**0.0079 ± 0.0057**	88.20 ± 0.61	0.0563 ± 0.0082	**0.0077 ± 0.0041**	70.06 ± 5.28	0.0151 ± 0.0118†	0.0109 ± 0.0073†	67.33 ± 3.06	0.0122 ± 0.0110†	0.0106 ± 0.0054†	84.12 ± 1.66	0.0278 ± 0.0121	**0.0079 ± 0.0064**
FLGMM	89.40 ± 1.02	0.0401 ± 0.0069	0.1081 ± 0.0074	89.81 ± 0.84	0.0454 ± 0.0082	0.1175 ± 0.0050	89.47 ± 0.98	0.0424 ± 0.0074	0.1120 ± 0.0083	89.48 ± 0.97	0.0423 ± 0.0080	0.1120 ± 0.0082	89.63 ± 0.87	0.0465 ± 0.0062	0.1153 ± 0.0058

non-IID (alpha=5)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	88.09 ± 4.01	0.0457 ± 0.0107	0.1050 ± 0.0215	89.91 ± 1.39	0.0485 ± 0.0043	0.1190 ± 0.0117	75.58 ± 13.92	0.0560 ± 0.1128†	0.0552 ± 0.0923†	75.31 ± 13.93	0.0510 ± 0.0968†	0.0603 ± 0.0801†	85.69 ± 2.65	0.0273 ± 0.0165	0.0748 ± 0.0260
FairFed	89.76 ± 0.83	0.0380 ± 0.0054	0.1095 ± 0.0074	89.92 ± 0.92	0.0448 ± 0.0073	0.1158 ± 0.0073	72.23 ± 12.29	0.0688 ± 0.0881†	0.0676 ± 0.0922†	67.94 ± 12.28	0.1327 ± 0.1254†	0.1180 ± 0.1131†	86.50 ± 2.31	0.0318 ± 0.0137	0.0838 ± 0.0214
Median	86.40 ± 1.26	0.0510 ± 0.0066	0.0965 ± 0.0102	86.51 ± 1.31	0.0607 ± 0.0078	0.1074 ± 0.0100	84.54 ± 1.36	0.0425 ± 0.0125	0.0800 ± 0.0120	84.59 ± 1.39	0.0448 ± 0.0144	0.0833 ± 0.0130	85.54 ± 1.32	0.0503 ± 0.0118	0.0930 ± 0.0091
FLTrust	89.77 ± 0.56	0.0450 ± 0.0049	0.1142 ± 0.0055	89.87 ± 0.72	0.0488 ± 0.0056	0.1194 ± 0.0046	**89.55 ± 0.70**	0.0441 ± 0.0066	0.1124 ± 0.0041	**89.55 ± 0.70**	0.0441 ± 0.0066	0.1124 ± 0.0041	89.68 ± 0.74	0.0481 ± 0.0051	0.1161 ± 0.0049
FedAA-DDPG	88.67 ± 1.19	0.0464 ± 0.0130	0.1072 ± 0.0159	88.73 ± 1.13	0.0490 ± 0.0140	0.1117 ± 0.0106	88.74 ± 1.21	0.0451 ± 0.0223	0.1077 ± 0.0166	88.89 ± 1.17	0.0421 ± 0.0123	0.1056 ± 0.0100	88.46 ± 1.18	0.0606 ± 0.0198	0.1198 ± 0.0199
LASA	89.46 ± 0.63	0.0379 ± 0.0056	0.1078 ± 0.0085	89.68 ± 0.72	0.0485 ± 0.0090	0.1179 ± 0.0062	86.33 ± 1.10	0.0301 ± 0.0074	0.0782 ± 0.0127	86.32 ± 1.16	0.0350 ± 0.0118	0.0818 ± 0.0101	88.34 ± 1.13	0.0409 ± 0.0028	0.1033 ± 0.0075
FairGuard	89.52 ± 1.09	0.0301 ± 0.0039	0.1023 ± 0.0091	89.95 ± 0.71	0.0380 ± 0.0099	0.1104 ± 0.0094	61.24 ± 16.96	0.0073 ± 0.0097†	0.0188 ± 0.0259†	62.72 ± 15.82	0.0119 ± 0.0166†	0.0226 ± 0.0329†	77.66 ± 15.66	0.0188 ± 0.0084†	0.0523 ± 0.0233†
FLTrust+FairGuard	**90.20 ± 0.46**	0.0359 ± 0.0058	0.1114 ± 0.0045	**90.17 ± 1.38**	0.0415 ± 0.0055	0.1154 ± 0.0087	81.41 ± 12.25	0.0407 ± 0.0225	0.0859 ± 0.0370	79.86 ± 12.64	0.0371 ± 0.0275†	0.0780 ± 0.0177†	78.76 ± 14.45	0.0404 ± 0.0131†	0.0882 ± 0.0291†
GuardFed-AD2+	88.59 ± 1.24	**0.0063 ± 0.0039**	0.0638 ± 0.0086	88.50 ± 0.76	**0.0104 ± 0.0073**	0.0579 ± 0.0090	88.39 ± 0.97	**0.0083 ± 0.0057**	**0.0609 ± 0.0148**	88.24 ± 0.82	**0.0119 ± 0.0095**	**0.0605 ± 0.0188**	88.47 ± 0.94	**0.0079 ± 0.0065**	0.0613 ± 0.0144
LoGoFair-DP	86.76 ± 3.64	0.0488 ± 0.0200	**0.0053 ± 0.0054**	88.32 ± 1.28	0.0567 ± 0.0113	**0.0060 ± 0.0038**	75.65 ± 13.47	0.0202 ± 0.0152†	0.0077 ± 0.0052†	75.26 ± 13.37	0.0218 ± 0.0174†	0.0082 ± 0.0059†	84.71 ± 2.33	0.0356 ± 0.0130	**0.0044 ± 0.0041**
FLGMM	89.18 ± 0.81	0.0392 ± 0.0068	0.1080 ± 0.0077	89.40 ± 0.95	0.0514 ± 0.0072	0.1185 ± 0.0072	89.38 ± 0.56	0.0422 ± 0.0047	0.1099 ± 0.0044	89.40 ± 0.96	0.0417 ± 0.0048	0.1094 ± 0.0105	**89.77 ± 0.62**	0.0434 ± 0.0079	0.1147 ± 0.0056

Reading: Higher ACC and lower AEOD/ASPD are better. Best/second-best eligible displayed means are **bold**/underlined; displayed ties share rank. † retains a disparity whose corresponding mean ACC is <=80%, excluding it only from fairness ranking. ACC ranking and all seeds/means/sample SDs are unchanged. Eligibility uses unrounded mean ACC; SD does not rank. This post-hoc display rule was author-approved on 2026-10-12, not preregistered. Marks do not establish significance or undefined metrics. ACC is percent; gaps are [0,1]; AEOD is the absolute TPR gap, not full equalized odds. Shading identifies GuardFed.

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Coverage and provenance: Eleven methods, two distributions, five conditions and 10 seeds give 1100 evaluation records, not 1100 independent training models: LoGoFair reuses FedAvg checkpoints. Backbone checkpoints are terminal round 70. FLGMM contributes 96 new and 4 explicitly reused screen records; its selected recipe is warmup 20/control width 2.0/local learning rate 0.001. Original mixed training/replay environments and separate CPU saved-state verification remain disclosed; no cross-runtime equivalence is claimed.

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CelebA - eleven-method native validation comparison

Appendix sensitivity analysis | 6 shared seeds: 91005-91010 | mean +/- sample SD (ddof=1)

Best: bold. Second-best: underlined. Fairness ranking requires mean ACC >80%; dagger marks ineligible values. Displayed ties share rank.

IID (alpha=5000)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	87.99 ± 2.11	0.0406 ± 0.0131	0.0992 ± 0.0184	89.71 ± 0.80	0.0535 ± 0.0119	0.1227 ± 0.0063	69.36 ± 6.14	0.0852 ± 0.0377†	0.0866 ± 0.0413†	67.41 ± 3.17	0.0823 ± 0.0434†	0.0698 ± 0.0304†	85.70 ± 1.38	0.0280 ± 0.0074	0.0738 ± 0.0142
FairFed	89.39 ± 0.89	0.0414 ± 0.0096	0.1090 ± 0.0082	89.65 ± 0.75	0.0480 ± 0.0047	0.1170 ± 0.0044	70.49 ± 3.83	0.0912 ± 0.0604†	0.0961 ± 0.0536†	65.00 ± 4.85	0.1179 ± 0.0658†	0.1133 ± 0.0622†	85.87 ± 1.25	0.0305 ± 0.0060	0.0790 ± 0.0076
Median	86.37 ± 1.01	0.0504 ± 0.0088	0.0977 ± 0.0107	86.58 ± 1.26	0.0602 ± 0.0102	0.1054 ± 0.0109	84.48 ± 0.83	0.0402 ± 0.0154	0.0786 ± 0.0124	84.83 ± 0.80	0.0440 ± 0.0083	0.0807 ± 0.0088	85.45 ± 1.23	0.0526 ± 0.0099	0.0956 ± 0.0101
FLTrust	89.41 ± 0.86	0.0392 ± 0.0056	0.1087 ± 0.0063	89.73 ± 0.92	0.0488 ± 0.0070	0.1189 ± 0.0046	**89.61 ± 0.88**	0.0385 ± 0.0059	0.1088 ± 0.0048	**89.61 ± 0.88**	0.0385 ± 0.0059	0.1088 ± 0.0048	**89.67 ± 0.71**	0.0475 ± 0.0063	0.1151 ± 0.0041
FedAA-DDPG	88.18 ± 1.27	0.0424 ± 0.0143	0.1053 ± 0.0090	89.00 ± 0.86	0.0431 ± 0.0131	0.1082 ± 0.0096	88.36 ± 1.04	0.0398 ± 0.0135	0.1015 ± 0.0091	88.53 ± 1.41	0.0337 ± 0.0137	0.0988 ± 0.0180	88.40 ± 0.55	0.0482 ± 0.0082	0.1108 ± 0.0096
LASA	89.52 ± 0.91	0.0374 ± 0.0060	0.1078 ± 0.0033	89.27 ± 0.69	0.0478 ± 0.0083	0.1158 ± 0.0048	85.62 ± 1.12	0.0257 ± 0.0069	0.0732 ± 0.0112	85.64 ± 1.70	0.0352 ± 0.0106	0.0811 ± 0.0151	87.87 ± 1.39	0.0449 ± 0.0092	0.1022 ± 0.0128
FairGuard	89.01 ± 0.92	0.0308 ± 0.0028	0.0969 ± 0.0048	89.28 ± 1.48	0.0372 ± 0.0093	0.1062 ± 0.0111	49.97 ± 1.80	0.0069 ± 0.0170†	0.0082 ± 0.0201†	51.67 ± 0.00	0.0000 ± 0.0000†	0.0000 ± 0.0000†	82.74 ± 2.15	0.0183 ± 0.0090	0.0446 ± 0.0234
FLTrust+FairGuard	**89.99 ± 0.65**	0.0359 ± 0.0077	0.1073 ± 0.0050	**90.32 ± 0.71**	0.0377 ± 0.0062	0.1132 ± 0.0096	79.32 ± 12.97	0.0223 ± 0.0112†	0.0615 ± 0.0351†	82.14 ± 14.30	0.0321 ± 0.0091	0.0800 ± 0.0345	88.87 ± 1.34	0.0422 ± 0.0103	0.1050 ± 0.0104
GuardFed-AD2+	88.56 ± 0.68	**0.0114 ± 0.0073**	0.0665 ± 0.0159	88.29 ± 0.75	**0.0148 ± 0.0104**	0.0581 ± 0.0129	88.57 ± 0.87	**0.0055 ± 0.0036**	**0.0697 ± 0.0110**	88.75 ± 0.79	**0.0070 ± 0.0040**	**0.0617 ± 0.0085**	88.35 ± 0.87	**0.0171 ± 0.0083**	0.0559 ± 0.0203
LoGoFair-DP	86.82 ± 1.92	0.0481 ± 0.0120	**0.0076 ± 0.0060**	88.22 ± 0.75	0.0541 ± 0.0086	**0.0099 ± 0.0030**	70.21 ± 6.01	0.0152 ± 0.0121†	0.0099 ± 0.0081†	68.05 ± 3.01	0.0133 ± 0.0136†	0.0101 ± 0.0041†	84.88 ± 1.23	0.0281 ± 0.0120	**0.0096 ± 0.0068**
FLGMM	89.12 ± 1.17	0.0411 ± 0.0082	0.1064 ± 0.0083	89.66 ± 1.00	0.0436 ± 0.0093	0.1162 ± 0.0044	89.10 ± 1.01	0.0431 ± 0.0092	0.1091 ± 0.0089	89.12 ± 1.01	0.0429 ± 0.0098	0.1091 ± 0.0088	89.32 ± 0.92	0.0460 ± 0.0073	0.1121 ± 0.0044

non-IID (alpha=5)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	86.90 ± 4.54	0.0464 ± 0.0125	0.0977 ± 0.0234	89.58 ± 1.62	0.0479 ± 0.0036	0.1151 ± 0.0128	78.23 ± 12.19	0.0756 ± 0.1376†	0.0788 ± 0.1078†	78.69 ± 12.03	0.0652 ± 0.1170†	0.0804 ± 0.0927†	86.13 ± 3.06	0.0312 ± 0.0166	0.0817 ± 0.0283
FairFed	89.56 ± 0.96	0.0388 ± 0.0033	0.1082 ± 0.0085	89.73 ± 1.11	0.0431 ± 0.0055	0.1133 ± 0.0064	75.25 ± 10.47	0.0727 ± 0.1009†	0.0770 ± 0.1069†	70.49 ± 12.37	0.0952 ± 0.1115†	0.0879 ± 0.1083†	86.17 ± 2.80	0.0299 ± 0.0162	0.0814 ± 0.0256
Median	86.19 ± 1.06	0.0525 ± 0.0042	0.0969 ± 0.0033	86.20 ± 1.30	0.0579 ± 0.0076	0.1044 ± 0.0095	84.49 ± 1.15	0.0423 ± 0.0100	0.0786 ± 0.0089	84.57 ± 1.19	0.0433 ± 0.0112	0.0817 ± 0.0094	85.68 ± 1.19	0.0508 ± 0.0143	0.0950 ± 0.0072
FLTrust	89.60 ± 0.61	0.0428 ± 0.0041	0.1114 ± 0.0041	89.75 ± 0.87	0.0482 ± 0.0064	0.1177 ± 0.0047	**89.48 ± 0.63**	0.0435 ± 0.0048	0.1103 ± 0.0030	89.48 ± 0.63	0.0435 ± 0.0048	0.1103 ± 0.0030	**89.66 ± 0.82**	0.0473 ± 0.0034	0.1151 ± 0.0058
FedAA-DDPG	88.52 ± 1.46	0.0452 ± 0.0150	0.1050 ± 0.0192	88.52 ± 1.31	0.0508 ± 0.0168	0.1108 ± 0.0130	88.88 ± 1.39	0.0367 ± 0.0061	0.1031 ± 0.0134	88.76 ± 1.28	0.0416 ± 0.0118	0.1035 ± 0.0113	88.29 ± 1.37	0.0564 ± 0.0186	0.1148 ± 0.0213
LASA	89.25 ± 0.66	0.0369 ± 0.0054	0.1048 ± 0.0087	89.61 ± 0.88	0.0476 ± 0.0076	0.1158 ± 0.0046	86.06 ± 1.23	0.0281 ± 0.0075	0.0751 ± 0.0137	86.04 ± 1.29	0.0307 ± 0.0020	0.0779 ± 0.0075	88.11 ± 1.33	0.0404 ± 0.0030	0.1011 ± 0.0082
FairGuard	89.09 ± 1.11	0.0305 ± 0.0043	0.0996 ± 0.0103	89.78 ± 0.80	0.0353 ± 0.0094	0.1063 ± 0.0074	61.06 ± 17.25	0.0069 ± 0.0083†	0.0170 ± 0.0214†	63.21 ± 15.26	0.0144 ± 0.0191†	0.0215 ± 0.0317†	79.49 ± 15.09	0.0182 ± 0.0102†	0.0594 ± 0.0181†
FLTrust+FairGuard	**90.39 ± 0.39**	0.0351 ± 0.0063	0.1122 ± 0.0054	**90.12 ± 1.74**	0.0417 ± 0.0069	0.1152 ± 0.0109	84.46 ± 5.99	0.0462 ± 0.0214	0.0966 ± 0.0220	84.73 ± 3.40	0.0320 ± 0.0093	0.0757 ± 0.0155	84.66 ± 9.74	0.0433 ± 0.0137	0.1013 ± 0.0128
GuardFed-AD2+	88.58 ± 1.42	**0.0062 ± 0.0039**	0.0641 ± 0.0067	88.43 ± 0.87	**0.0110 ± 0.0086**	0.0570 ± 0.0090	88.49 ± 0.88	**0.0083 ± 0.0044**	**0.0566 ± 0.0081**	88.42 ± 0.83	**0.0067 ± 0.0068**	**0.0611 ± 0.0114**	88.42 ± 1.10	**0.0089 ± 0.0080**	0.0583 ± 0.0172
LoGoFair-DP	85.68 ± 4.12	0.0411 ± 0.0201	**0.0061 ± 0.0063**	88.02 ± 1.50	0.0547 ± 0.0129	**0.0066 ± 0.0046**	78.31 ± 11.31	0.0263 ± 0.0141†	0.0077 ± 0.0061†	78.54 ± 11.02	0.0306 ± 0.0141†	0.0085 ± 0.0065†	85.07 ± 2.69	0.0344 ± 0.0159	**0.0056 ± 0.0045**
FLGMM	89.05 ± 0.99	0.0398 ± 0.0039	0.1067 ± 0.0079	89.33 ± 1.13	0.0508 ± 0.0055	0.1169 ± 0.0076	89.34 ± 0.65	0.0403 ± 0.0047	0.1081 ± 0.0020	**89.63 ± 0.67**	0.0418 ± 0.0027	0.1112 ± 0.0070	89.64 ± 0.73	0.0444 ± 0.0087	0.1143 ± 0.0062

Reading: Higher ACC and lower AEOD/ASPD are better. Best/second-best eligible displayed means are **bold**/underlined; displayed ties share rank. † retains a disparity whose corresponding mean ACC is <=80%, excluding it only from fairness ranking. ACC ranking and all seeds/means/sample SDs are unchanged. Eligibility uses unrounded mean ACC; SD does not rank. This post-hoc display rule was author-approved on 2026-10-12, not preregistered. Marks do not establish significance or undefined metrics. ACC is percent; gaps are [0,1]; AEOD is the absolute TPR gap, not full equalized odds. Shading identifies GuardFed.

Method scope: Native predictions preserve each method's own postprocessing; this does not isolate aggregation effects. FairFed/FairGuard/composition, FedAA and LASA are documented adaptations. LoGoFair-DP uses fitted demographic-parity postprocessing, 20 image-ID virtual cohorts (not training clients), fit seed 1719 and 30 postprocessing rounds. FLGMM uses the author-code full-model/largest-cluster adaptation and native uncalibrated outputs.

Coverage and provenance: Eleven methods, two distributions, five conditions and 10 seeds give 1100 evaluation records, not 1100 independent training models: LoGoFair reuses FedAvg checkpoints. Backbone checkpoints are terminal round 70. FLGMM contributes 96 new and 4 explicitly reused screen records; its selected recipe is warmup 20/control width 2.0/local learning rate 0.001. Original mixed training/replay environments and separate CPU saved-state verification remain disclosed; no cross-runtime equivalence is claimed.

Selection boundary: Seed 91001 participated in validation selection; other validation seeds and historical test metadata were exposed. The 9/6-seed panels are sensitivity views, not untouched test cohorts. No score ranking, significance, final-test or complete 17-method claim is made. This display does not select the manuscript's final primary endpoint.

Source: root-adopted eleven-method native1100 display table; no new statistics, fits, inference or training.